SULTAN IDRIS EDUCATION UNIVERSITY

DEP 3023: TEACHING & INSTRUCTIONAL MODEL

SEMESTER 2 SESSION 2024/2025

GROUP PROJECT

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GROUP A

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1.0 Introduction

Environmental awareness is an important element of sustainable living, as it promotes responsible behavior to preserve the natural world. In today's world, environmental issues such as pollution and climate change are becoming more serious. One of the most effective and practical ways to help reduce environmental damage is to properly sort and dispose of waste. A trash sorter refers to a system that allows for segregation of trash and recyclables into separate containers. Unfortunately, many people, especially children and young students lack the knowledge to correctly separate different types of the waste.

The Trash Sorter game is a simple and interactive drag-and-drop educational game designed to teach users, particularly young students, how to sort waste into three main categories which are organic, non-organic and hazardous waste. The game is designed to be user-friendly to make it easier for users to play it. It presents various types of trash items and users must drag and drop each item into the correct bin. Through this interactive approach, users will receive immediate feedback on their choices and provide corrections for them so that they can learn from their mistakes. Additionally, by using visual, sound and reward systems the learning process can be more enjoyable. This game will encourage users to apply their knowledge from the game in real- time situations.

2.0 Background Studies

Waste management remains one of the most challenging environmental problems globally. The problem tends to get worse due to the lack of awareness and proper education about handling waste from a young age. Educating a generation about the environment requires the introduction of waste education at an early stage.

Education through games has worked efficiently for learners, especially children who find interacting with fun activities enticing. Games provide the best atmosphere for relaxed learning that reduces monotonous cognitive fatigue while boosting motivation and curiosity in children, which is not always possible in traditional settings with heavy theoretical training, resulting in tedium.

An instance of technology used for environmental education is the Android-based "Sorting Waste Game" developed by Rahmayanti et al., which teaches children how to classify waste into the various categories of organic, inorganic, general, hazardous, and more while sorting them accordingly. By gamifying the process with appropriate visuals and instant feedback, the application fosters fundamental learning in children while promoting proactive environmentalism.

Building a smart, sustainable society requires shifting environmental education to the foundation level and using digital technologies like mobile apps and websites relevant to today's youth.

3.0 Problem Statement

Nowadays, many students lack knowledge of how to correctly dispose of different types of waste, leading to improper sorting and increased pollution. Traditional teaching methods, such as lectures or textbook-based lessons, often fail to fully engage young students and help them retain important knowledge about waste management. The large scale of education programs such as workshops or physical learning can be costly and may not be sustainable for all schools. The financial limitation reduces the effort to effective environmental education. There is a need for a more impactful method to instill environmental responsibility in students.

4.0 Learning goal

The primary learning goals of the Trash Sorter game are:

- To educate and empower students to become environmentally responsible by teaching them proper waste management through engaging gameplay.
- To encourage familiarity with digital tools in environmental learning.
- To enhance decision-making skills through learning experiences.

5.0 Learning objective

By playing the Trash Sorter game, users will be able to :

- Identify and categorize different types of waste.
- Understand the environmental impact of improper waste disposal.
- Apply their knowledge in real-life situations to help reduce pollution.

6.0 Scope

The scope of the Trash Sorter game project is described by a framework to be compatible with good game design practices. It covers all the fundamentals to provide an immersive and meaningful digital courseware experience to learn.

Core Idea

Trash Sorter is an environmental education activity that educates about the process of selecting and discarding between different categories of household waste. It applies an interactive drag and drop feature for sorting items in three categories: Organic, Non-Organic, and Hazardous Waste. The game aims to improve awareness of the environment, as well as decision-making, through an interesting and exciting activity.

Platform and Technical Specifications

Platform: Web-based

Tools: Scratch

Device compatibility: Works on PCs, tablets, and Chromebooks

Game Features

- Simple drag and drop interface for ease of use
- Has visual representation of waste items (apple, paper, syringe) and disposable bins with distinct colours (organic waste, non-organic waste, and hazardous waste)
- A scoring system to give points for correct sorting
- “Game Over” prompt pop-up for mistakes

- Sound effects (“ding” for correct answers and siren sound for false answers)

Game limitation

- No multiplayer
- One main level (endless mode with randomized items)
- No mobile app

7.0 Significance of the project

This project is significant because it introduces a scalable and cost-effective alternative to traditional approaches. The Trash Sorter game, a digital technology that allows schools to deliver a quality environmental education on computers, tablets or smartphones. The project highlights the importance of integrating digital tools in schools. The basic aim of learning through games is to lead students to learn through interactive engagement, mistakes and repetition, in a manner that leads them to become engaged in the game, without being aware of the fact that they are also learning at the same time. In teaching practice, teachers can use educational digital play as a motivational introduction to a new content, as an additional activity, in order to encourage student interaction with content, as well as motivation for learning. Research confirms that serious games greatly contribute to increasing motivation for learning and they encourage student efficiency and effectiveness in learning (Rajic & Tasevska, 2019). In addition, the game encourages critical thinking and responsible behavior to help students understand the real-time situation. This project supports sustainable education practices by making environmental learning interactive, affordable and accessible for a wide range of students.

8.0 Instructional System Design Model

To make the Trash Sorter game facilitate environmental education for kids, we utilized the ADDIE model, which is made up of Analysis, Design, Development, Implementation, and Evaluation, using Scratch as the development platform.

1. Analysis

As for the educational concern, children's awareness of waste disposal and recycling is often minimal. The intended users are learners from 7 to 10 years old, and they learn best through images and direct interaction. The analysis also verified that the game-based approach is optimal for capturing and sustaining attention while reinforcing understanding of the different classes of waste.

2. Design

The game was designed with the following primary learning goals in mind:

- Identify and classify various waste types (hazardous, non-hazardous, and organic).
- Apply their knowledge in real-life situations to help reduce pollution.

A drag-and-drop system with Scratch allows students to sort trash into designated bins. To keep players interested, the game has audio effects and vibrant graphics. Players can play the single main level indefinitely

3. Development

The essential game features include:

- Drag-and-drop mechanics
- Score tracking for correct/incorrect sorting
- Game over pop-ups for wrong answers
- Animated sprites for bins and waste items

4. Implementation

A web browser can be used to access the game, which was made available through the Scratch website. The game can be used at home or in the classroom by both teachers and students. The cloud sharing capabilities of Scratch make distribution simple and installation-free.

5. Evaluation

Feedback from educators and student user testing was used to conduct the formative evaluation. Scratch's integrated scoring system and event logs aid in evaluating student performance. The effectiveness of the game in enhancing students' comprehension of waste management was assessed using post-play reflections and feedback forms.

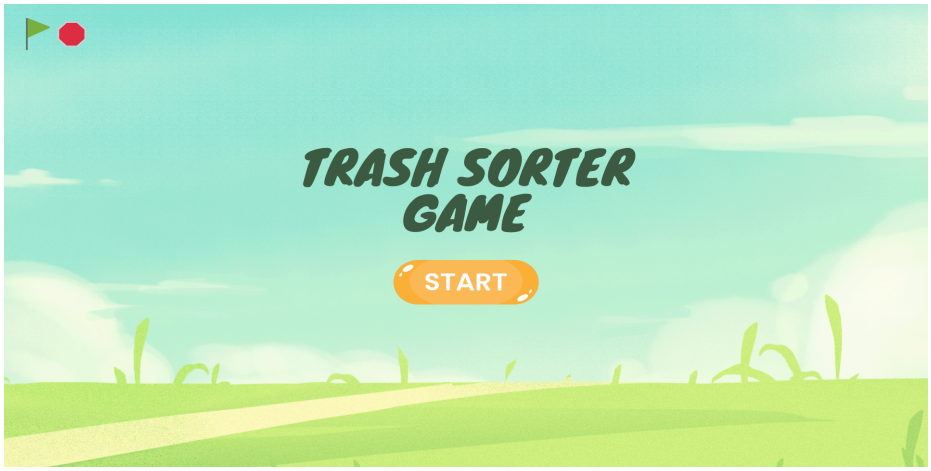
9.0 Learning theory

Trash Sorter game is built based on two key theories, constructivism and Behaviorism. These approaches help the development of an interactive and impactful educational experience that can enhance knowledge and behavior change. Constructivism emphasizes active learning, where students build their own understanding of the world through hands-on experiences and interactions. In educational games, this means providing opportunities for students to experiment, explore, and discover solutions to problems within the game environment. Users engage directly with the content by drag-and-drop waste items into bins, they construct their understanding through trial and error and receive immediate feedback on their action. Mistakes are used as a learning process that help users to improve their waste-sorting skills. The game encourages critical thinking and allows young students to use the knowledge they get from the game in real life situations.

Behaviorism focuses on observable behavior and the effects of external stimuli on learning. In the Trash Sorter game, Users will earn points on rewards when they sort waste items correctly. They can repeat the game until they sort it correctly and build a long-term memory of proper waste classification. This is the effective approach for building habits through consistent feedback.

10.0 Storyboard

Scene 1



- Main page for the Trash Sorter Game
- Player must click the “Start” button to start the game

Scene 2

- Three recycling bins are shown, which represent organic, non-organic and hazardous waste in a distinct colour
- Start/stop button and score are at the top left of the game

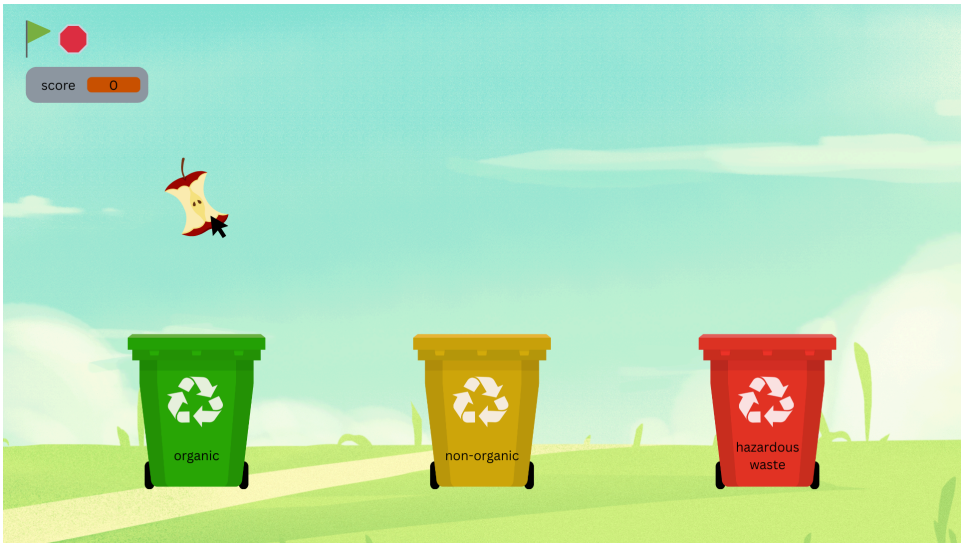
Scene 3

- Random waste object will spawn and fall downward (eg: apple)

Scene 4

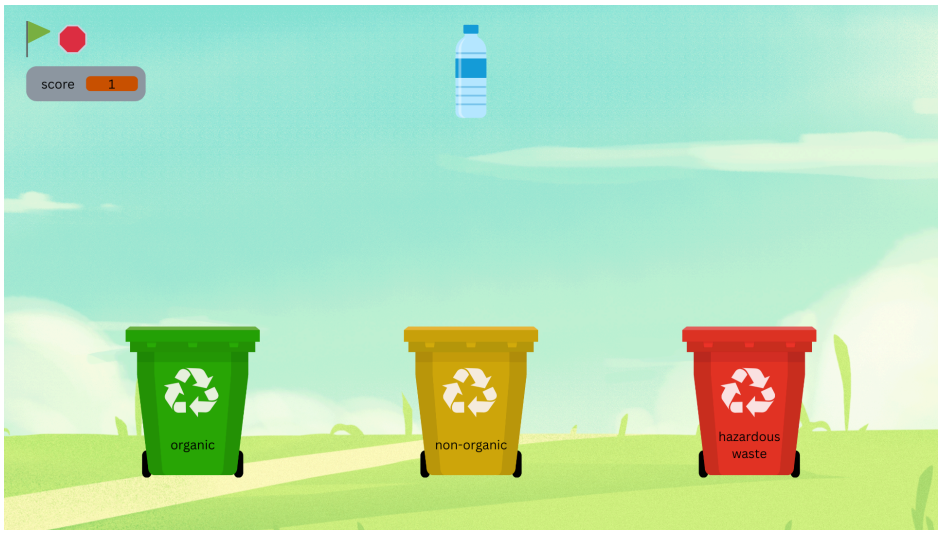
- Place the cursor at the particular waste

Scene 5



- Drag and place at the correct recycling bins (eg : apple for organic waste)

Scene 6



- The score will be rewarded if the match is correct
- Next random waste object will be spawned

Scene 7

- If the player mismatches the waste object (eg : plastic bottle is a non-organic waste)

Scene 8

- The game over prompt will appear, and the player has to restart the game

11.0 References

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5. Adipat, S., Laksana, K., Busayanon, K., Asawasowan, A., & Adipat, B. (2021). Engaging students in the learning process with game-based learning: The fundamental concepts. International Journal of Technology in Education (IJTE), 4(3), 542-552. https://doi.org/10.46328/ijte.169